



REPUBLIC OF KENYA

Ministry of Health

Basic Obstetric Protocols

2026

1st Edition



Ministry of Health 2026

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Foreword

The Kenya Government through the Ministry of Health has made remarkable progress in improving maternal and newborn health outcomes through various interventions though more investment must be put in place to attain the Sustainable Development Goals. The Kenyan constitution 2010 Article 43 1(a), Article 28 and the Health Policy Framework (2014-2030) goal is to offer the highest quality attainable standard of health and be responsive to the needs of the Kenyan citizens. Continuous improvement on quality maternal, newborn, child and adolescent health services is a priority for the Kenyan Government that strives to in cooperate /employ evidence based best practices and actionable interventions by use of timely obstetric protocols.

These Basic Obstetric Protocols 2026 have been developed to provide evidence based, practical guidance to health care professionals at all levels of health service delivery. The instrument addresses the Big Seven priority conditions in maternal care namely, Antepartum Haemorrhage, Postpartum Haemorrhage, Obstructed/Prolonged Labor, Anaemia in Pregnancy, Hypertensive Disorders in Pregnancy, Sepsis, and Abortion complications which together account for majority of the maternal and perinatal morbidity and mortality in our country. The protocols outlined herein reflect the current evidence on best practices in clinical recommendations adapted to our national context, as guided by the National Guidelines on Quality Obstetrics and Perinatal Care, 2024. They are designed for use by health care professionals to ensure standardized, timely, quality health care. Target population for these obstetric protocols include National and County health managers, Health Policy Makers, Regulators, Health Care Providers, Researchers, Academia, and Pre-service training institutions. We recommend all healthcare professionals to familiarize themselves with these obstetric protocols and to apply them consistently in their daily practice. Routine use and application of these protocols will significantly reduce preventable adverse outcomes and contribute to safer childbirth for all women.

It is my sincere hope that this document which is designed to provide a practical guide and equip health care professionals with maternal health knowledge, skills, competencies and positive attitude at all levels of health service delivery implementation; will accelerate the reduction of maternal and perinatal morbidity and mortality. Together, we can uphold our commitment to ensuring that every pregnancy and every birth is safe.



Dr. Bashir Issak
Director of Family Health
Ministry of Health

Acknowledgment

The development of the Basic Obstetrics Protocols 2026 involved in-depth consultations with a wide range of multidisciplinary stakeholders including specialized and experienced health care professionals through literature review, consultative meetings and alignment to the National Guidelines on Quality Obstetrics and Perinatal Care 2024.

The Division of Reproductive Maternal Newborn, Child and Adolescent Health wish to express its profound gratitude to all individuals and organizations whose contributions made the development of this Obstetric Protocols possible. The DRMNCAH is indebted to the team of stakeholders constituting the technical working group both at National and County Levels including experts from Moi Teaching and Referral Hospital and Kenyatta National Hospital.

We also acknowledge the role of professional bodies such as Kenya Obstetric and Gynaecology Society (KOGS), Midwifery Association of Kenya (MAK), medical training institutions, regulatory bodies, both development and implementing partners. We extend our heartfelt gratitude to Clinton Health Access Initiative (CHAI) for their financial and technical support..

Our appreciation also goes to the esteemed reproductive health coordinators, midwives, nurses, medical officers, obstetricians and gynaecologists, and other frontline health care professionals who shared their real-world experiences and practical insights. Your voices ensured that the protocols are not only clinically sound but also realistic for use in diverse health service delivery settings. It is our sincere hope that these protocols will serve as a trusted companion for healthcare professionals across all levels of the health system and thus ensure that every woman has the opportunity for a safe and positive birth experience.



Dr. Edward Serem

Head, Division of Reproductive, Maternal, Newborn, Child and Adolescent Health

Ministry of Health

Appreciation

The Ministry of Health extends profound gratitude to the Kenya Obstetrical and Gynaecological Society (KOGS) for their visionary leadership in synthesizing the 1st Edition of Basic Obstetric Protocols. This landmark achievement serves as a vital cornerstone for strengthening our health system, through evidence-based standardization that is essential for high-quality maternal care, particularly during life-saving obstetric emergencies. The KOGS has shepherded an indispensable instrument that reinforces the knowledge acquired in training to practice, directly contributing to the reduction of preventable mortality and ensuring a safer future for mothers and newborns across the nation.

Authors

	Name	Organization
1	Dr. Kireki Omanwa	KOGS
2	Dr. Mandeep Sura	KOGS
3	Prof. Moses Obimbo	KOGS
4	Dr. Maina Ngoigo	KOGAC
5	Dr. Laurn Mengesa	KOGS
6	Dr. Hiba Tajihud	KOGS

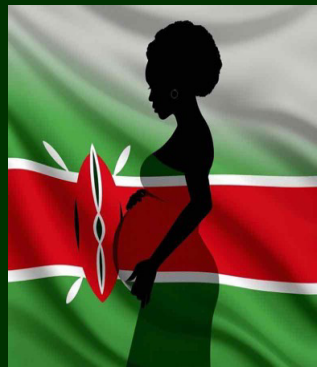
Contributors

	Name	Organization
1	Dr. Serem Edward	MOH
2	Dr. Janette Karimi	MOH
3	Dr. Loise Nyanjau	MOH
4	Dr. Juliet Omwoha	MOH
5	Rosemary Kihoto	CHAI
6	Dr. Philomena Owende	KNH
7	Dr. Mutindi Kakuti	KOGS
8	Scholastica Wabwire	MOH-RMNCAH
9	Hellen Mutsi	MOH
10	Janet Mogire	DRMNCAH-MOH
11	Ian Ochieng	KOGS
12	Dr. Oketch Dan	MSE-IC

13	Grace Asami	MAK
14	Prof. Alfred Osoti	UON
15	Dr. Rita Arogo	WHO
16	Leonard Tin	Independent
17	Dr. Hilda Gikunda	CHAI
18	Dr. Grace Watau	Nairobi county
19	Alinda Ndenga	Lwala
20	Eunice Atsali	MAK/KU
21	Dr. George Gwako	UON
22	Charles Rombo	NPHI
23	Dr. Mary Maina	Kiambu County
24	Sylvia Ayitso	MOH
25	Dr. Judith Awinja	MOH
26	Ruth Muia	MOH-NCD

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Glossary

Abbreviation	Full name
ANC	Antenatal Care
ANP	Antenatal Profile
APH	Antepartum Haemorrhage
APLS	Antiphospholipid Syndrome
ARB	Angiotensin Receptor Blocker
AST	Aspartate Aminotransferase
ALT	Alanine Aminotransferase
ACE	Angiotensin Converting Enzyme
ABCDE	Airway, Breathing, Circulation, Disability, Exposure
BP	Blood Pressure
BPP	Biophysical Profile
BTU	Blood Transfusion Unit
BD	Twice Daily
bpm	Beats per Minute
BTU	Blood Transfusion Unit
CBC	Complete Blood Count
CNS	Central Nervous System
CPR	Cardiopulmonary Resuscitation
CRP	C-Reactive Protein
CT	Computed Tomography
CTG	Cardiotocography

Abbreviation	Full name
CCT	Controlled Cord Traction
DIC	Disseminated Intravascular Coagulation
DQA	Data Quality Assessment
FHR	Foetal Heart Rate
FHG	Full Haemogram
GCS	Glasgow Coma Scale
GXM	Group and Cross Match
Hb	Hemoglobin
HDU	High Dependency Unit
HELLP	Hemolysis, Elevated Liver enzymes, Low Platelets
HIV	Human Immunodeficiency Virus
HR	Heart Rate
ICU	Intensive Care Unit
IM	Intramuscular
INR	International Normalized Ratio
IU	International Units
IUFD	Intrauterine Fetal Death
IPTp-SP	Intermittent Preventive Treatment in pregnancy with Sulfadoxine-Pyrimethamine
IV	Intravenous
LDH	Lactate Dehydrogenase
LFTs	Liver Function Tests

Abbreviation	Full name
MAP	Mean Arterial Pressure
MCH	Mean Corpuscular Hemoglobin
MCHC	Mean Corpuscular Hemoglobin Concentration
MCV	Mean Corpuscular Volume
MEOWS	Modified Early Obstetric Warning Signs
MgSO ₄	Magnesium Sulphate
MRI	Magnetic Resonance Imaging
MVA	Manual Vacuum Aspiration
NASG	Non-Pneumatic Anti-Shock Garment
NBU	Newborn Unit
NEWS	National Early Warning Score
NRM	Non-Rebreather Mask
NSAIDs	Non-Steroidal Anti-Inflammatory Drugs
NRFS	Non-Reassuring Fetal Status
OD	Once Daily
OR	Operating Room
POC	Point of Care
OPOCUS	Obstetric Point of Care Ultrasound
PPH	Postpartum Haemorrhage

Abbreviation	Full name
PPROM	Pre-labour Premature Rupture of Membranes
PPE	Personal Protective Equipment
PR	Pulse Rate
PV	Per Vaginal
QID	Four Times a Day
RBC	Red Blood Cells
RR	Respiratory Rate
SBP	Systolic Blood Pressure
SIRS	Systemic Inflammatory Response Syndrome
SpO ₂	Peripheral Oxygen Saturation
SLE	Systemic Lupus Erythematosus
STIs	Sexually Transmitted Infections
STAT	Immediately
SVD	Spontaneous Vaginal Delivery
TDS	Three Times Daily
TXA	Tranexamic Acid
UEC	Urea, Electrolytes, Creatinine
UHC	Universal Health Coverage
WHO	World Health Organization

Key Considerations: Obstetric red flags and referral pathways

Take a comprehensive history to identify High risk patients such as:

Pre-existing and/or newly diagnosed health conditions:

High blood pressure, diabetes, anaemia, heart disease, kidney disease, thyroid disorders, autoimmune diseases (like lupus), obesity, or infectious diseases (HIV, hepatitis).

Extremes of maternal age:

Age is younger than 20 years old or older than 35 years old.

Multiple gestation:

Twins, triplets, or more increase the risks for both mother and babies.

Past pregnancy complications:

History of preterm birth, recurrent pregnancy loss, previous caesarean birth and other surgeries, Pre-eclampsia, PPH, APH, obstructed labor, still birth, or having a child with a genetic or birth defect.

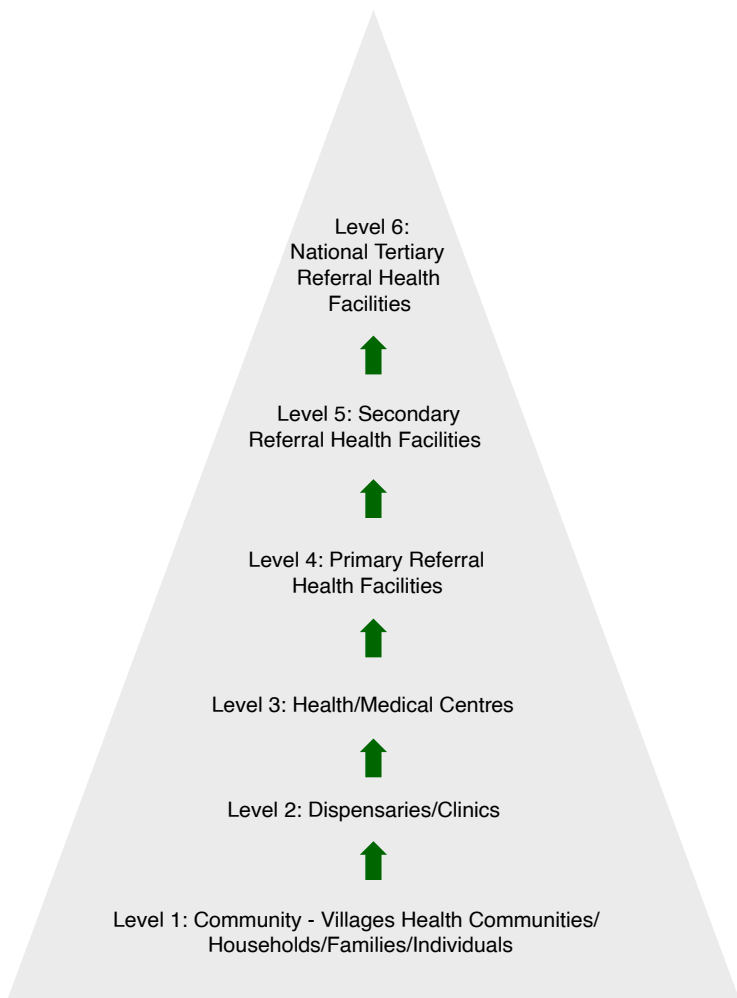
Lifestyle factors:

Use of alcohol, tobacco, illegal drugs, malnutrition, GBV or poor health seeking behavior including inadequate prenatal care.

Fetal Factors:

Babies diagnosed with genetic conditions and/or birth defects.

Should the patient exhibit any of the above, the health professional in lower tier facilities should refer the client to a level 4 OR above facility for further assessment.



1. Hypertensive disorders in pregnancy

Classification of Hypertensive disorders of pregnancy

Chronic hypertension.	Hypertension confirmed preconception or prior to 20 weeks' gestation.
Gestation hypertension.	New onset hypertension arising after 20 weeks' gestation in the absence of proteinuria and other systemic symptoms.
Pre-eclampsia.	Multi-systemic disorder of unknown aetiology characterized by elevated BP $\geq 140/90$ developed after 20 weeks of gestation in a previously normotensive woman.
Pre-eclampsia superimposed on chronic hypertension.	Features of Pre-eclampsia developing in a woman who had hypertension prior to conception.
Eclampsia.	Occurrence of one or more generalized tonic clonic seizures superimposed on Pre-eclampsia. Seizures may occur at any point during pregnancy and within the 12 weeks postpartum period.
HELLP syndrome *Although not a classification of HDP, it is an important complication spectrum of Pre-eclampsia.	HELLP syndrome describes a constellation of: • Hemolysis, • Elevated Liver enzymes showing hepatic dysfunction (AST and ALT greater than twice their normal range.), • Low Platelets (platelet count of less than 100,000/dl). It is associated with maternal mortality and severe morbidity including disseminated intra-vascular coagulopathy, liver hematoma, liver failure, and renal failure.

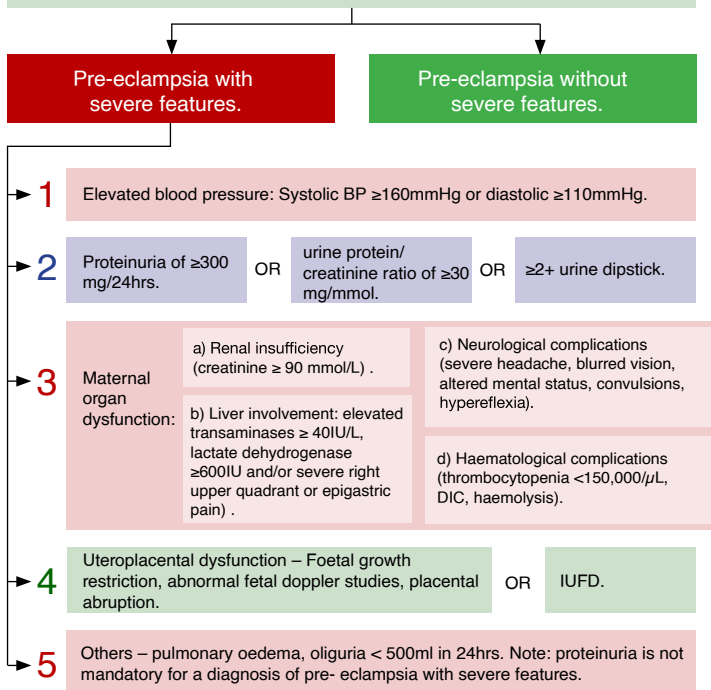
Pre-eclampsia

Definition of Pre-eclampsia

Multi systemic disorder of unknown aetiology characterized by elevated systolic $\geq 140\text{mmHg}$ and/or Diastolic $\geq 90\text{mmHg}$ developed after 20 weeks of gestation, and upto 12 weeks postpartum, in a previously normotensive woman.

Classification of Pre-eclampsia

Pre-eclampsia is classified with severe features and without severe features:



Risk factors:

- Nulliparity.
- Age: <20 years or >35 years.
- Chronic hypertension.
- Chronic kidney disease.
- Autoimmune diseases (APLS, SLE).
- Inherited Thrombophilias.
- Inter-pregnancy interval > 10 years.
- Diabetes mellitus.
- Family history of Pre-eclampsia.
- Multi-fetal gestation.
- Previous history of Pre-eclampsia.
- Vitamin D deficiency.
- Obesity.

Antenatal care

- Elicit risk factors during preconception care, ANC visits or at first contact with the patient and refer appropriately.
- Educate the patient on danger signs and symptoms. Schedule two weekly follow up visits for full assessment of maternal and fetal well being.
- Use of low-dose aspirin can reduce the risk of preterm Pre-eclampsia by over 62% when started prior to 16 weeks of gestation.
- Initiate 150mg aspirin once a day, and refer to level 4 hospital or above. Ideally initiate at 11-14 weeks of gestation (or at first contact after that) until 36 weeks.
- Initiate Calcium 1g OD at 12 weeks or earliest contact thereafter.

Early Assessment	
History	Ask about headaches, visual disturbances, epigastric pain, and chest pain. Note any signs of mental confusion or agitated behaviour. Ask about fetal movements.
Physical examination	Examine for pallor, jaundice, dehydration, oedema. Vitals: Blood pressure. Heart rate. SpO2. Systemic examination: • Cardiovascular: presence or absence of any abnormal heart sounds. • Respiratory: Auscultate for air entry, comment on normal or abnormal. Presence or absence of crepitations. • Abdominal: Right upper quadrant or epigastric tenderness. Do Leopold maneuver. • CNS: GCS and reflexes.
Laboratory investigations	FHG, UEC, LFT, urinalysis, uric acid, LDH.
Imaging	Obstetric ultrasound. If indicated, CT/MRI brain.

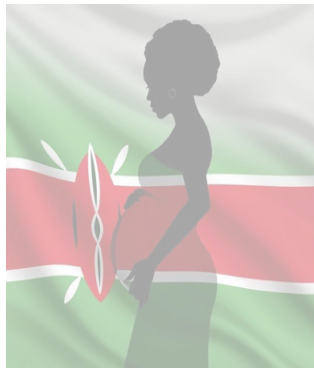
Principles of management

Pre-eclampsia without severe features



- Schedule 2 weekly follow up visits for serial BP measurements. If BP is equal to or above 160/100 or patient develops any signs of severity, refer to a hospital with an obstetrician.
- Educate the patient on self monitoring, danger signs and symptoms, weight management, exercise and good diet. Involve nutritionist. Discuss the birth plan.
- If BP is $\geq 150/90$, initiate antihypertensive treatment.
- The goal is to maintain diastolic blood pressure between 80 and 100mm Hg and systolic pressure between 130 and 150mm Hg.
- First-line medications are:
 - labetalol initial dose 100mg BD. Maximum daily dose is 2400mg.
 - Nifedipine 20mg BD. Maximum daily dose of 80mg.
 - Methyldopa 500mg TDS. Maximum daily dose of 3000mg.

Note: Atenolol, ACE inhibitors, ARBs, and diuretics should be avoided during pregnancy.



Pre-eclampsia with severe features

- 1 Admit patient.
- 2 Blood pressure control.
- 3 Evaluate for end organ damage. Do base line investigations.
- 4 Seizure prophylaxis.
- 5 Fluid management: 60-80ml/h.
- 6 Fetal surveillance: CTG, BPP. Dexamethasone if < 34 weeks.
- 7 Develop a delivery plan: Timing and mode of delivery.

↓
Assess the need for referral to a higher level facility and ensure thorough documentation.

↓
Assess for conservative management versus delivery

≥37+0 weeks	Delivery indicated regardless of symptoms. SVD preferred unless obstetric indication.
34 to 36 ⁺⁶ weeks	Delivery may be indicated in the event of development of worsening signs and symptoms.
<34 weeks	Delivery indicated in the setting of fetal/maternal compromise and in the setting of an NBU.

Management of severe acutely elevated blood pressure.

Target blood pressure	• Target sBP range of 130 to 150mmHg. • Target dBP range 80 to 90mmHg. Aim for gradual and sustained lowering of BP so blood flow to the fetus is not compromised.
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Anti-hypertensives for use in severe acute elevated blood pressure.

Drug	Initial dose	Follow-up
Labetalol	20mg IV gradually over 2 minutes.	Repeat BP measurement at 10-minute intervals: • If BP remains above target level at 10 minutes, give 40mg IV over 2 minutes. • If BP remains above target level at 20 minutes, give 80mg IV over 2 minutes. • If BP remains above target level at 30 minutes, give 80mg IV over 2 minutes. Cumulative maximum dose is 300mg. If target BP is not achieved, move to another agent. Hold dose if HR<60 beats per minute.
OR		
Hydralazine	5mg IV gradually over 1 to 2 minutes.	Repeat BP measurement at 20-minute intervals: • If BP remains above target level at 20 minutes, give 5mg IV over 2 minutes, depending on the initial response. Cumulative maximum dose is 30mg per treatment event. If target BP is not achieved, switch to another class of agent.

Postpartum considerations

- Ensure continuous monitoring of blood pressure and maintenance within the normal range.
- Ensure completion of magnesium sulphate till 24 hours post delivery.
- Labetalol, nifedipine, hydralazine, amlodipine, atenolol, metoprolol, captopril and enalapril can be used in breastfeeding.
- Methyldopa should be avoided due to risks of postpartum depression.
- Women should be educated on the role of Pre-eclampsia prophylaxis in subsequent pregnancies and the long- term cardiovascular risk.
- Advise to initiate preconception care when planning next pregnancy and after conception early enrollment at high-risk antenatal clinic.
- Advise on appropriate family planning method.

Eclampsia

Described as occurrence of one or more generalized tonic clonic seizures. Seizures may occur at any point during pregnancy including 12 week post partum period.

General principles of management

1. SHOUT FOR HELP: Urgently mobilize available personnel. Rapid assessment of mother's condition.
 - Danger: Ensure you are safe, put on appropriate PPE. Ensure the environment is safe.
 - Response: Mobilize, call for help.
 - Airway: Open the mother's airway using a head tilt, chin-lift, or jaw thrust maneuver. Secure the airway using oropharyngeal airway if available.
 - Breathing: Confirm presence and rate, if present put on oxygen via NRM at 15 L/min in recovery position.
 - Circulation: Check presence and adequacy of pulse. If absent or <60bpm, commence CPR with 30 chest compressions, then give two rescue breaths. Ensure IV access, take blood samples for FBC/LFTS/UEC/coag/GXM.
2. Blood pressure control. (Refer to table on Anti-hypertensives for use in severe acute elevated blood pressure).
3. Evaluate capacity to manage patient and initiate early referral if necessary.
4. Active seizures should be treated with IV. Magnesium sulphate as first line agent. Sign's of Magnesium toxicity should be monitored. (Refer to table below).
5. Fluid management.

Magnesium sulphate: Agent of choice to control and prevent recurrent seizures.

Magnesium sulphate	Magnesium sulphate MUST be diluted to a 20% solution for IV use.
Preparation of 4g 20% solution	• Wash hands thoroughly with soap and running water or use 70% alcohol hand rub and air dry. For an available solution of 50% MgSO₄: • Using a 20-ml syringe, draw 12mL of sterile water for injection. • Withdraw 8mL of MgSO₄ 50% solution* and add to the 12 ml of water for injection to make 20mL of 20% solution. (4g per 20 ml).

Administration of loading dose and maintenance dose.	Pritchard regimen(IV/IM): Loading dose (IV and IM): • Give 4g IV over five minutes (20mL of the prepared 20% magnesium sulfate solution). • Ensure aseptic technique when giving magnesium sulfate deep IM injection. Warn the woman that she will have a feeling of warmth when the magnesium sulfate is given. • Follow promptly with 10g of 50% magnesium sulfate solution: Give 5g (10mL of the undiluted 50% solution) in each buttock as a deep IM injection with 1 ml of 2% lidocaine in the same syringe. • If convulsions recur after 15 minutes, give 2g (10mL of the prepared 20% magnesium sulfate solution intravenously over 5 minutes. Maintenance dose (Intramuscular): Give 5g (10mL of the undiluted 50% magnesium sulfate solution) with 1mL of 2% lidocaine in the same syringe by deep IM injection into alternate buttocks every four hours. Continue treatment for 24 hours after birth or the last convulsion, whichever occurs last. Zuspan regimen (IV/IV): Intravenous administration (use an infusion pump if available): Loading dose: • Give 4g IV over five minutes (20mL of the diluted 20% magnesium sulfate solution). • If convulsions recur after 15 minutes, give 2g (10mL of the diluted 20% magnesium sulfate solution) IV over 5 minutes. Maintenance dose (intravenous): • Give intravenous infusion 1g (5mL of the diluted 20% magnesium sulfate solution) per hour. • Continue treatment for 24 hours after childbirth or the last convulsion, whichever occurs last.
Monitor for signs of toxicity.	• Count respiration rate for 1 minute every hour. The rate should be ≥ 16. • Patella reflexes should be present. Check every hour. • Insert an indwelling urinary catheter and measure urinary output hourly. Output should be $\geq 30\text{ml/hour}$. <i>Note: loss of patellar reflex is the first sign of Magnesium toxicity.</i>

Antidote.	The antidote (calcium gluconate) should be kept ready. In case of respiratory arrest: • Assist ventilation (mask and bag, anaesthesia apparatus, intubation). • Give calcium gluconate 1g (10ml of 10% solution) intravenously slowly over 3 minutes, until respiration begins to counteract the effect of magnesium sulfate.
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Management of suspected magnesium toxicity

Signs and symptoms	Physical examination and laboratory investigations	Management
Reduced urine output (<0.5ml/kg/H for over 4 hours).	Hydration status. Respiratory examination. Review fluid balance. Check deep tendon, reflexes, respiratory rate, heart rate. Send renal profile and serum Mg if available.	If tendon reflex is ABSENT, STOP MgSO₄ infusion. If tendon reflex is present, adjust infusion rate.
Absence of deep tendon reflex.	Check respiratory rate and heart rate. Send renal profile and serum Mg if available.	Stop MgSO₄ infusion. Perform ECG for evidence of prolonged PR interval or wide QRS complex. Consider restarting infusion at lower dose when reflexes return.
Respiratory depression.	Check respiratory rate and heart rate. Send renal profile and serum Mg if available.	Stop MgSO₄ infusion. Maintain airway, put patient in recovery position. Consider giving IV calcium gluconate.
Respiratory arrest.	Assessment is per maternal collapse (ABC). Send renal profile and serum Mg if available.	Stop MgSO₄ infusion. Maintain airway with intubation and ventilation. Give IV calcium gluconate.

Note: Therapeutic level of serum Mg – 1.7-3.5 mmol/L.

Fluid Management

General principles of fluid management are as below:

General principles of fluid management:	1. Total fluid given to women with severe hypertension (60-80ml/hour) of crystalloid). 2. Strict input and output charting. 3. Fluid challenge: when indicated, should be done with careful assessment of the woman's hydration status and after consultation with a specialist. 4. Oral fluid restriction is not routinely practiced except in cases of proven fluid overload, i.e., in acute pulmonary oedema. Women with fluid overload should be managed in a high dependency unit/intensive care unit with co-management from anaesthetist and physician. 5. Diuretics should NOT be given in the event of oliguria unless there is evidence of acute pulmonary oedema.
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Management of Eclampsia

Loading dose MgSO₄

4g IV over 20 mins via controlled infusion.

If seizures occur/ongoing while preparing MgSO₄/re-occurs while giving MgSO₄

- Diazepam 5-10mg IV at rate of 2-5mg/min (max (10mg).
- Clonazepam 1-2mg IV over 2-5mins.
- Midazolam 5-10mg IV over 2-5mins/ IM.
- Maintenance doses of MgSO₄.
 - 1g per hour via controlled infusion for 24 hours after birth or after last convulsion then review requirement.

If impaired renal function

- Reduce maintenance doses to 0.5g per hour and consider serum monitoring.

Monitor:

- BP and pulse every 5mins until stable then every 30mins.
- Respiratory Rate and patellar reflexes hourly.
- Temperature 2 hourly.
- Urine output hourly.
- Strict fluid balance monitoring.
- Therapeutic serum management level 1.7 to 3.5mmol/liter.
- Stop infusion if toxicity is clinically suspected.

Review with consultant if:

- Urine output <80mls in 4 hours.
- Deep tendon reflexes absent and RR<12 breaths/min.

ANTIDOTE

- Calcium Gluconate 1g intravenously slowly over 3 minutes.

Control seizures

Control Hypertension

If ante-partum plan delivery ASAP

TREAT HYPERTENSION

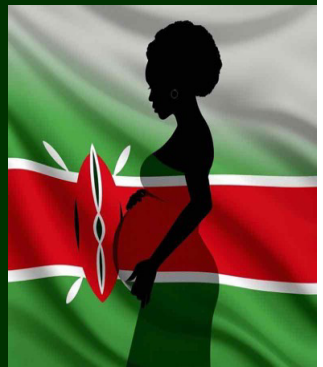
- If systolic BP $\geq$ 160mmHg or diastolic BP $\geq$ 100mmHg.
- Aim to reduce Systolic BP to 130-150mmHg and diastolic BP to 80-100mmHg.
- Avoid maternal hypotension.
- Monitor fetal heart rate with continuous CTG.

ANTIHYPERTENSIVE MEDICATION

- PO Nifedipine 10-20mg, repeat after 45 mins, max 80mg.
- Hydralazine: 5-10mg IV over 3 to 10 mins. Repeated doses 5mg IV every 20mins if required. Max dose 30mg. It may also be given as a 10-20mg IV infusion and titrated against BP response.
- Labetalol: Initially 20mg IV bolus over 2mins. Repeated doses 40-80mg IV every 10mins to max of 300mg if required.

BIRTH

- Continue close fetal monitoring.
- Stabilize mother prior to birth.
- Deliver within 8-12 hours after onset of seizures.



2. Obstetric Haemorrhage

Antepartum Haemorrhage (APH)

Definition

Bleeding after 24 weeks of gestation and before delivery.

Differentials	Relevant History
• Placenta previa. • Placental abruption. • Uterine rupture. • Vasa Previa. • Cervical or Vaginal lesions. • Trauma.	1. Gestational age by date and ultrasound). 2. Findings of prior ultrasounds (placentation? Presentation?). 3. Duration of signs and symptoms. 4. Foetal status – perceives foetal movements? frequency?. 5. Any clinical symptoms suggestive of shock/ hypo-tension such as general weakness/malaise, dizziness, fatigues, palpitations, diaphoresis. 6. Abdominal pain.

Maternal - General Evaluation	Maternal - Physical Examination	Foetal Evaluation
Signs of shock: • Abnormal BP <90/60mmHg or >140/90 mmHg. • Weak or rapid pulse • Tachycardia >100 bpm. • Tachypnoea >20 bpm. • Temperature < 36.1°C or >37.2°C. • Decreased urine output <30ml/hr. • Altered mental state, including reduced alertness and awareness, confusion, and sleepiness. • Cold, moist skin. Hands and feet may be pale.	DO NOT DO digital vaginal exam till ultrasound excludes previa. 1. Uterine palpation, size, tenderness, soft/hard?. 2. Speculum – a speculum examination is mandatory to evaluate for possible causes of bleeding and exclude differentials (listed above).	1. Presence of foetal heart rate (CAUTION: Differentiate from mothers' pulse as may be tachycardic!). 2. Close FHR monitoring in >28/40 – Pinnards/ Doppler/CTG. 3. Normal FHR 110-160 BPM and regular. 4. OPOCUS should be performed immediately where available to assess viability of foetus, placentation and presentation.

Management

SHOUT FOR HELP!!! To urgently mobilize nurses, obstetrician on call, midwives, theatre, radiology and anesthesia teams.

Danger – Put on appropriate PPE.

Response – Mobilise, call for help.

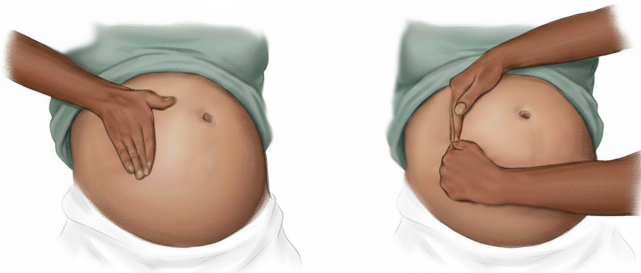
Circulation – Fix two wide bore cannulas. Do full hemogram and Group and cross match blood. Start IV crystalloids(ringers lactate or normal saline). If pulse is absent or <60bpm, start CPR immediately aiming at 100 high quality (achieving a depth of at least 2 inches, and allow for full chest recoil) compressions per minute.

Airway – Open mother's airway using a head-tilt chin-lift or jaw thrust manoeuvre. Secure the airway using oropharyngeal airway if available.

Breathing – Look, listen and feel for breathing. Put on oxygen at 10L/min.

NB: Ensure left lateral tilt of the uterus.

Perform secondary survey after initial maternal stabilization.

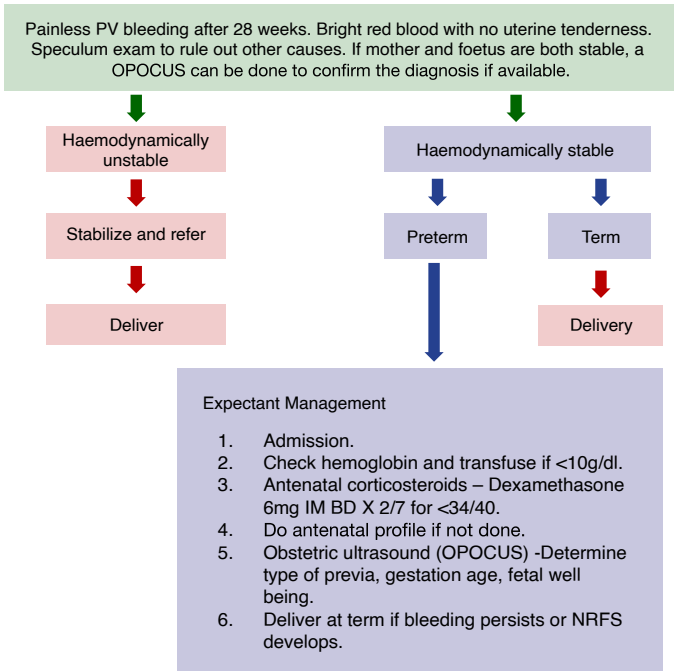


Placenta Previa

Definition

- Placenta Previa refers to the implantation of the placenta over or near the internal os of the cervix.
- Presents with painless bleeding.

Algorithm for management of Placenta Previa



NB: Early recognition, immediate stabilization of patient and urgent transfer to a facility with a theatre, BTU and a consultant obstetrician for further management.

Placenta Abruption

Premature separation of the placenta, presents with uterine tenderness and non reassuring fetal heart rate pattern.

Vaginal bleeding can be obvious or concealed.

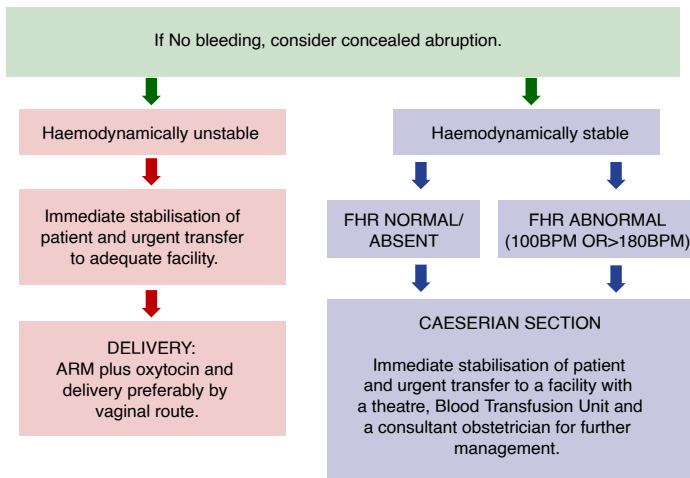
OPOCUS has ruled out placenta previa.

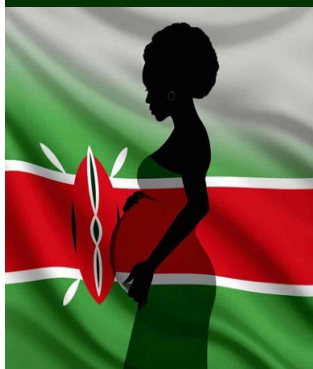
Types

Revealed: amount of visible blood loss commensurate with level of shock

Concealed: amount of external blood loss disproportionate with features of anaemia or shock.

Algorithm for management of Placenta Abruption





Post Partum Haemorrhage

Definition

- Blood loss of $\geq 500\text{mls}$ OR $\geq 300\text{mls}$ plus abnormal clinical signs and symptoms after vaginal delivery.
- Blood loss of $\geq 1000\text{mls}$ or more after cesarean delivery.

Classification

Primary postpartum hemorrhage: Primary (immediate) PPH occurs within the first 24 hours after delivery.

Secondary postpartum hemorrhage: Secondary (late) PPH occurs between 24 hours after delivery of the infant and up to 6 weeks postpartum. Most late PPH is due to retained products of conception, infection, or both.

Etiology

The 4 "T"s:

- **Tone (70%): uterine atony.**
- **Trauma (20%): uterine, cervical, or vaginal injury, uterine inversion.**
- **Tissue (10%): retained placenta (cotyledon or succenturiate lobe) or clots.**
- **Thrombopathy (<1%): pre-existing or acquired coagulopathy.**

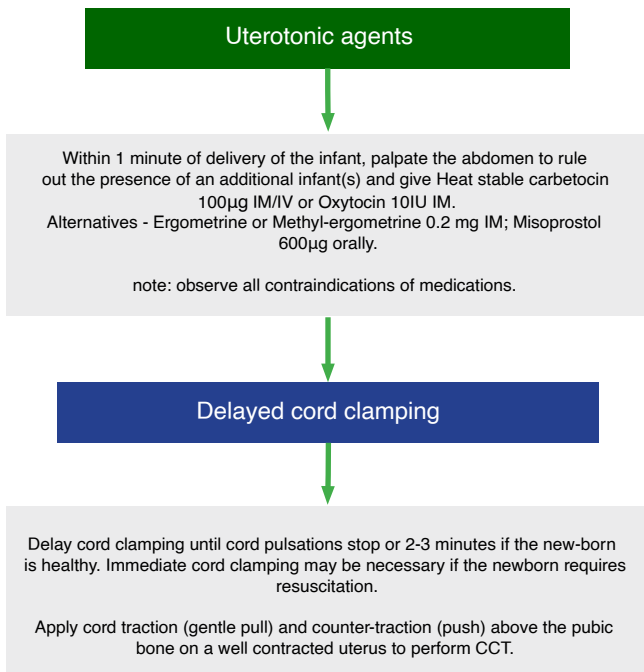
More than one cause may however be present e.g., tone and tissue.

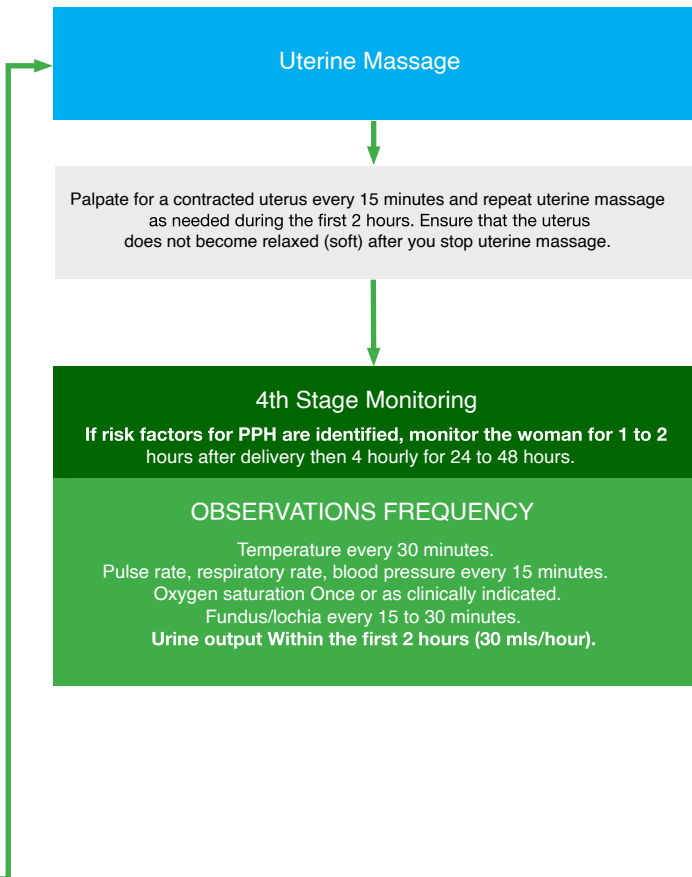
Prevention of PPH

1. Preconception care and Antenatal care

The WHO recommends daily iron and folic acid supplementation with 30mg to 60mg of elemental iron and 400µg (0.4 mg) of folic acid. This should be started as soon as pregnancy is confirmed or at preconception.

2. Active management of third stage of labor





Management of PPH

The EMOTIVE mnemonic refers to a comprehensive strategy for treating PPH. It is a structured approach to manage Postpartum Hemorrhage (PH) efficiently in emergency settings.

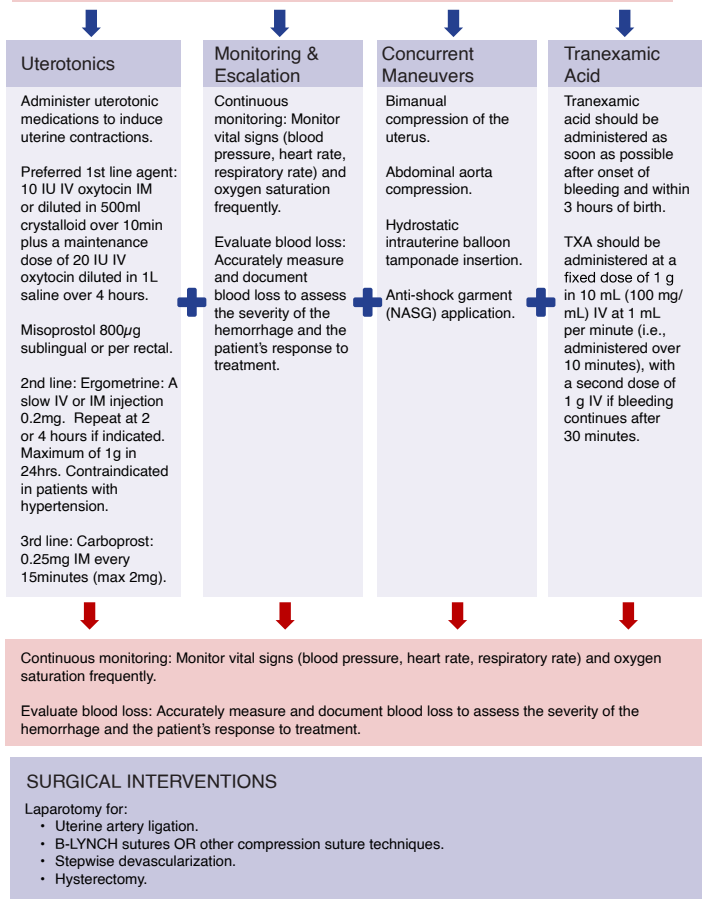
E	M	O	T	IV	E
Early detection and trigger criteria	Massage of uterus	Oxytocic drugs	TXA	IV fluids	Examination and escalation
Calibrated drape for the collection of blood with trigger lines at 300 ml and 500ml for the first h after birth. Observations (blood loss, blood flow and uterine tone) every 15 min documented on the blood-loss monitoring chart. Blood pressure and pulse monitored once in the first hour postpartum and documented on the blood-loss monitoring chart. Trigger criteria: 1) clinical judgment. 2) blood loss 2500 ml. 3) blood loss 2300ml, plus one abnormal observation.	Massage until uterus has contracted or for 1 min.	10 IU IV oxytocin injection or diluted in 500 ml crystalloid over 10 min plus a maintenance dose of 20 IU IV oxytocin diluted in 1,000 ml saline over 4 h (with misoprostol 800 ug if used).	1g IV injection of TXA or diluted in 200 ml crystalloid over 10-min period.	IV fluids in addition to the infusion should be given if clinically indicated for resuscitation and will require a second IV access.	Ensure bladder is empty, evacuate clots and check for tears with an internal examination and placenta for completeness. Escalate if bleeding does not stop after first response or if clinician is unable to identify or manage cause of bleeding.
Implementation strategies Audit newsletters: sharing with all staff monthly detection and bundle use rates along with PPH, severe PPH, blood transfusion, laparotomy and death from PPH rates and given feedback at monthly departmental meetings. Champions: midwife and doctor to oversee change, troubleshoot, give feedback on audit newsletters and connect with other champions through chats, meetings and websites for sharing knowledge and lessons learned. Trolley or carry case: restocking of all medicines and devices used for PPH treatment after every use and completion of stocking checklist at the start of every shift. Training: on-site, simulation-based and peer-assisted lasting 90 min to an entire workday facilitated by provider guides, flipcharts and job aids displayed in labor wards.					

Management of ATONY

INITIAL RESPONSE

- Shout for help. A skilled health care team should be alerted immediately, midwives, obstetricians, anesthetists and blood bank personnel.
- Insert 2 wide bore cannula (gauge 16/18) to allow for fluid and/or blood administration.
- Take blood samples for grouping and cross matching and preparation of appropriate blood products.
- Set up IV infusion of either normal saline or ringer's lactate,
- Insert Foley's catheter to empty the urinary bladder and leave the catheter in situ.
- Perform fundal massage by gently massaging the uterus until it is firm and contracted.
- Administer oxygen by mask to maintain oxygen saturation.
- Evaluate capacity of your facility and consider need for transfer of the woman to the next level of care.

SIMULTANEOUS ACTIONS. TEAM LEAD SHOULD BE APPOINTED TO COORDINATE THESE INTERVENTIONS.



Tears

Vaginal/Cervical/Perineal tears

- Repair the identified tears and lacerations either in the ward or theatre.

Cervical tears

- Examination under anaesthesia (If required).
- Application of fundal pressure to enable close visualization of the cervix. §Use sponge/ring forceps to examine the cervix in its entirety.
- Repair identified cervical tears.
- NB: Cervical tears extending beyond the vaginal vault, laparotomy may be recommended.

Tissue

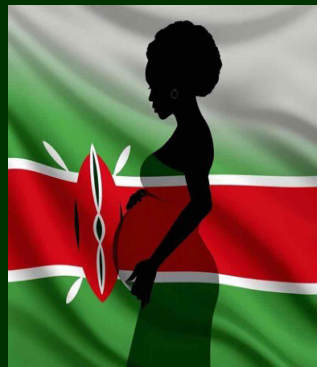
Retained placenta

- Double set-up is ideal, inform theatre to prepare, GXM.
- Empty bladder using self-retaining catheter.
- Give additional 10 IU of oxytocin.
- Apply controlled cord traction in a gentle fashion to avoid cord avulsion. If the placenta is still adherent, consider manual removal of the placenta.
- For retained fragments, uterine exploration by hand. Remove the fragments by hand, ovum forceps or wide curette. Examine the cervix, vagina and perineum for any tears and repair accordingly.

Thrombin

Use blood products to help control hemorrhage.

- Give fresh whole blood, if available, to replace clotting factors and red cells. If fresh whole blood is not available, choose one of the following based on availability:
 - Packed (or sedimented) red cells for red cell replacement.
 - Fresh frozen plasma for replacement of all clotting factors (15 ml/kg body weight).
 - Cryoprecipitate to replace fibrinogen and factor VIII in case of disseminated intravascular coagulation.
 - Platelet concentrates (if bleeding continues and the platelet count is less than 20,000).



3. Anemia in pregnancy

Definition:

Anemia is a condition in which the hemoglobin is $<11\text{g/dl}$.

Anemia:

Severity:	Haemoglobin values:
Mild	10 – 11 mg/dl.
Moderate	7 – 10 mg/dl.
Severe	$< 7\text{ mg/dl}$.

Etiology of Anemia during pregnancy:

No.	Classifications	Types
1	Physiological anemia.	
2	Pathological anemia.	• Iron deficiency. • Folic acid. • Vitamin B12 deficiency.
3	Haemolytic anemia.	• Familial - congenital jaundice, sickle cell anemia. • Acquired - malaria, severe infection.
4	Bone marrow insufficiency.	Hypoplasia or aplasia due to radiation, drugs, or severe infection.
5	Hemoglobinopathies.	Abnormal structure of one of the globin chains of the hemoglobin molecule ex-sickle cell disease.

Clinical features of Anemia:

Symptoms:

Fatigue, weakness, shortness of breath, low physical and mental capacity, headaches, dizziness, lightheadedness, awareness of heartbeat (palpitations), Pica (craving for non-food items like clay or ice).

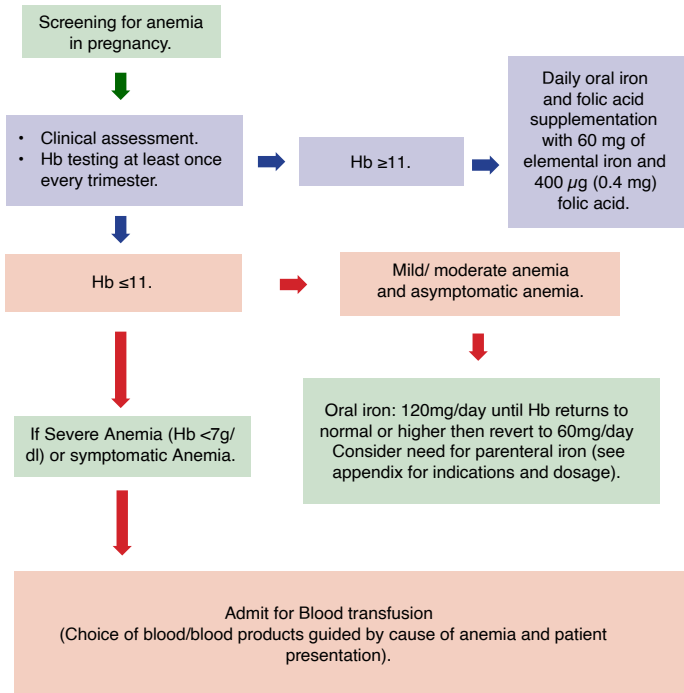
Signs:

Alopecia, mucosal pallor (conjunctival, tongue, nail bed, palmar), tachypnea, angular stomatitis, glossitis, koilonychia (spoon-shaped nails), fundal height smaller than gestational age, tachycardia, systolic murmurs and Congestive cardiac failure in severe cases, Jaundice, hepatosplenomegaly.

Diagnosis of Anemia in pregnancy:

Screening for Anemia in pregnancy	
Clinical Assessment – Symptoms and Signs.	
A comprehensive history must be taken.	
Hb testing at least once in every trimester, and during peripartum.	
Lab and imaging assessment (Guided by clinical findings)	
Complete Blood Count.	Hemoglobin (<11 g/dl) , hematocrit (<33%), Mean corpuscular volume. (MCV) between 80-100 is normal , mean corpuscular hemoglobin. (MCH) of 27-33 pg, mean corpuscular hemoglobin concentration. (MCHC) of 32-36 g/dl.
Reticulocyte count.	Normally 0.5-2.5% of RBC, if more than 5% could indicate hemolytic anemia.
Peripheral blood smear.	Presence of Microcytic/macrocyclic morphology, malaria parasites.
Liver Function tests (LFTs).	Elevated ALT, AST,LDH, bilirubin in HELLP syndrome, other hemolytic anemias and chronic liver disease.
Infectious causes.	Stool ova and cyst, occult blood in stool – hookworm, schistosomiasis. Blood also found in stool in case of gastric ulcers, hemorrhoids and other causes of upper and lower GI bleeding.

Management of Anemia in pregnancy:



Prevention of Anemia in Pregnancy:

In settings where anemia in pregnant women is a severe public health problem (40% of higher), a daily dose of 60 mg of elemental iron is preferred over a lower dose.

Preconception: :	Antenatal:
• Identify risk factors for iron deficiency. • Addressing non-nutritional causes of anemia in endemic pockets with special focus on malaria, hemoglobinopathies and fluorosis. • Hb and iron status assessed and, if deficient, appropriately treated before attempting conception. • Begin daily oral iron and folic acid supplementation with 30 mg to 60 mg of elemental iron and 400 µg (0.4 mg) folic acid for 3 months prior to conception. • Nutritional counselling - incorporate diet rich in iron, vitamin B12 and folate rich foods.	• Screening for anemia at least once in every trimester. • Daily oral iron and folic acid. supplementation with 30 mg to 60 mg of elemental iron and 400 µg (0.4 mg) folic acid is recommended. • Nutritional counselling - incorporate diet rich in iron, vitamin B12 and folate rich foods. • Preventive anthelmintic treatment in 2nd trimester with a single dose of mebendazole 500 mg or a single dose of albendazole 400mg. • For malaria endemic zones, Intermittent preventive treatment in pregnancy with sulfadoxine pyrimethamine (IPTp-SP) from 2nd trimester with doses one month apart for at least 3 doses.

Intrapartum:

Active management of 3rd stage of labour to prevent postpartum hemorrhage.

Postpartum:

- Avail modern methods of family planning to increase the inter-pregnancy interval.
- Nutritional counselling - incorporate diet rich in iron, vitamin B12 and folate rich foods.
- Supplementation of iron (60mg) and folic acid (400mg) up to 3 months postpartum.

4. Obstructed/Prolonged labour

Definition

Failure of the fetus to descend through the birth canal in spite of good uterine contractions, either from physical or mechanical obstruction.

Recommendations:

- Educate all mothers on importance of ANC attendance.
- Screen high risk individuals for gestational diabetes.
- Correct use and interpretation of labour monitoring tools for decision-making.

Etiology

Cephalopelvic disproportion

Occurs when there is a misfit between the foetal head and the pelvis. This means it is difficult or impossible for the fetus to pass safely through the pelvis. Cephalopelvic disproportion may be due to a small pelvis with a normal size head, or a normal pelvis with a large fetus, or a combination of a large baby and small pelvis.

Risk factors

Maternal

- Bony obstruction due to: contracted pelvis, tumours of pelvic bones, previous surgery or fractures of the pelvic bone.
- Soft tissue obstruction due to:
 1. Uterus: impacted subserous pedunculated fibroid and/or fibroids in the lower segment.
 2. Cervix: cervical dystocia.
 3. Vagina: septa, stenosis, tumours.
 4. Ovaries: Impacted ovarian tumours.

Foetal:

- Malpresentation and malposition's: e.g., Persistent occipital-posterior and deep transverse arrest, Persistent mento-posterior presentation, Brow, Shoulder dystocia.
- Large sized foetus (macrosomia). EFW of 4kg and above.
- Congenital anomalies: e.g., Hydrocephalus, Foetal ascites, Foetal tumours.
- Locked and conjoined twins.
- Any multiple gestation where first foetus is not cephalic.

Management

History

Examination

Signs of maternal distress on general examination include:

- Exhaustion.
- High temperature (38°C).
- Rapid pulse.
- Signs of dehydration: dry tongue and cracked lips.

Abdominal examination:

- The uterus is hard and tender. Presence of strong uterine contractions without relaxation.
- Fetal parts cannot be felt easily.
- Fetal heart rate is Absent or evidence of fetal distress.
- Rising retraction ring (Bandl's ring) is seen and felt as an oblique groove across the abdomen.

Vaginal examination:

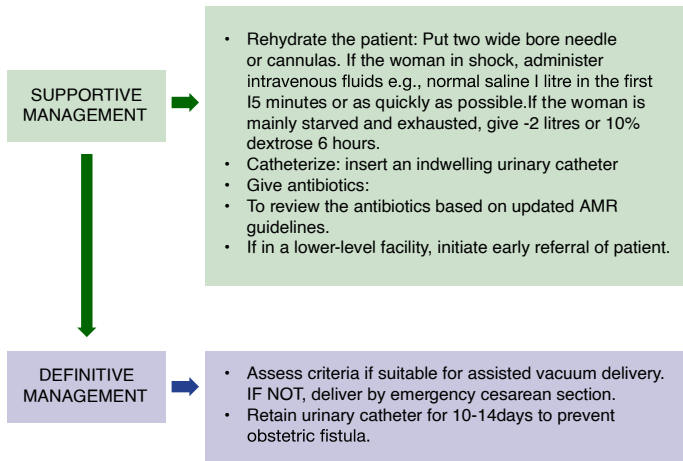
- Vulva is oedematous.
- Vagina is dry and hot.
- Cervix is fully or partially dilated, oedematous and hanging.
- The membranes are ruptured, presence of meconium stained, foul smelling liquor. Though based on the duration prior to presentation, liquor may be absent.
- The presenting part: is high and not engaged or impacted in the pelvis. If it is the head, it shows excessive moulding and large caput.

Differential diagnosis

- Constriction ring.
- Fundal myoma.
- Full bladder.

Complications	
Maternal:	Foetal:
• Maternal distress and ketoacidosis. • Rupture uterus. • Necrotic vesico-vaginal fistula. • Infections as chorioamnionitis and puerperal sepsis. • Postpartum haemorrhage due to injuries or uterine atony.	• Asphyxia. • Intracranial haemorrhage from excessive moulding. • Birth injuries. • Infections.

Treatment



Techniques for Impacted Head during cesarean section:

Vaginal Pushing (Abdominovaginal Method): An assistant inserts a hand into the vagina to push the fetal head upward while the surgeon extracts the baby.

Reverse Breech Extraction: The surgeon reaches into the uterus to grasp the feet, delivering the baby breech-first, which often eases delivery of the impacted head.

Uterine Incision Management: Due to the thinned lower segment and potential for extension, a low-segment incision may need extension (e.g., J or T-shaped) to avoid excessive trauma to the cervix or bladder.

5. Maternal sepsis:

Definition:

Life threatening condition defined by organ dysfunction resulting from infection during pregnancy, childbirth, post- abortion or within the postpartum period (42 days from delivery). It's the third leading cause of direct maternal mortality globally.

Puerperal sepsis is commonly due to endometritis.

Other infections include:

- Chorioamnionitis.
- Pyelonephritis.
- Genital tract infections.
- Puerperal sepsis.
- Malaria, viral and fungal diseases.
- Other infections during pregnancy such as HIV and STIs (gonorrhea, chlamydia, trichomonas).

Risk factors for maternal sepsis:

- Maternal comorbidities:
 - » Malnutrition.
 - » Obesity.
 - » Anemia.
 - » Endocrinopathies such as uncontrolled Diabetes Mellitus.
 - » Immunosuppression.
- Genital infections- Bacterial vaginosis, STIs.
- Group B streptococcus infection.
- Spontaneous events and/or interventions by care providers:
 - » Early rupture of membranes.
 - » Cervical cerclage.
 - » Frequent vaginal examinations.
 - » Manual removal of clots and placenta.
 - » Caesarean section.
 - » Non aseptic techniques.

Clinical picture

Symptoms	Clinical signs	Sites of infection
• Constant abdominal/ pelvic pain. • Foul vaginal discharge/ liquor. • Fevers and chills. • Diaphoresis. • Nausea & vomiting. • Lower urinary symptoms. • Productive cough • Generalized maculopapular rash. • Conjunctival suffusion.	• Temperature $>38^{\circ}\text{C}$ or $<36^{\circ}\text{C}$. • Tachycardia $>110\text{b/min}$. • Hypotension: SBP$<90\text{mmHg}$. • Tachypnea: RR$>24\text{br/min}$. • Low saturations $<90\%$ room air. • Fetal tachycardia. FHR$>160\text{b/min}$. • Pelvic/ uterine tenderness. • Renal angle tenderness. • Uterine sub-involution. • Capillary refill$>2\text{seconds}$. • Altered mental status, GCS <15. • Urine output $<0.5\text{ml/kg/hr}$.	• Skin and subcutaneous tissue. • Uterus. • Adnexa. • Cervix. • Vagina. • Perineum.

Diagnostic Scoring criteria for sepsis

Modified Early Obstetric Warning Score (MEOWS)

Score	3	2	1	0	1	2	3
Respiratory Rate (bpm)				9 - 18	19 - 25	26 - 30	>30
Pulse rate (bpm)		<40	40 - 50	51 - 100	101 - 110	111 - 129	>129
Temperature		<35	35-35.9	36 - 37.4	37.5 - 37.9	38.0 - 38.9	>39
Systolic blood pressure (mmHg)	<70	71 - 80	81 - 100	101 - 164	165 - 200	>200	
Diastolic blood pressure (mmHg)				<5	95 - 104	>105	
Conscious level	Unresponsive	Responds to pain	Responds to voice	Alert	Irritated		
Urine hourly (ml/h) or in 24h		>30 ($<720\text{ml}$)	>45 ($<1000\text{ml}$)	>45 ($>000\text{ml}$)			

Final score = Sum of individual scores at any one time.

Score	Action required.
0 or 1	Repeat observations when appropriate for clinical scenario.
2	Inform midwife in charge, repeat in 15 min.
3	Inform midwife in charge, obstetric registrar and duty anesthetist.
≥4	Consultant obstetrician, consultant anesthetist, and intensive care team should be informed.

Septic shock

Septic shock (below parameters despite adequate Intravenous fluid therapy)

- Sepsis+ need for vasopressors, SBPs <90mmhg.
- MAP less than 65mmhg.
- Lactate more than 2mmol/l.

Investigations

Laboratory investigations	Imaging tests
- Haemogram-white blood count >15,000/mm³ or <4,000mm³. - Urea, creatinine, electrolytes. - Liver function tests. - Blood culture. - Procalcitonin, C-reactive protein. - Blood slide for Malaria parasites. - Blood glucose. - Lactic acid levels. - Urine -urinalysis, urine culture. - High vaginal and endocervical swabs for M/C/S. - Coagulation abnormalities INR>1.5, APTT >60 sec. - Blood gas analysis.	- Pelvic ultrasound. - Abdominal xray. - Abdominal CT scan. - Chest Xray (shielded if pregnant). - Fetal surveillance- Obstetrics scan, CTG.

MANAGEMENT PRINCIPLES

1. Multi-disciplinary team approach as needed- involve infectious disease specialists, critical care team, general surgery teams alongside the team of obstetricians.
2. Early and timely referral as necessary after initial stabilization.
3. General principles- supportive care:
 - a. Regular monitoring of all vitals and recording in a MEOWS or NEWS chart.
 - b. Correction of hypoxia- Oxygen therapy.
 - c. Fluid support- give crystalloids as per need based on clinical assessment and blood pressure, pulse parameters.
 - d. Pain control and fever control-Acetaminophen, NSAIDs, opioids.
 - e. Empirical antibiotics- 'Golden hour rule'.
 - f. Correct anemia if present- Iron or transfusion.
 - g. Venous thromboembolism prophylaxis-Low molecular weight heparin.
 - h. Admit into intensive care unit (ICU) or high dependency Unit (HDU) if features of severe sepsis or septic shock or needing inotropic support.

“Golden hour rule” : In cases of maternal sepsis, initiate antibiotics within the first hour of presentation.

Give IV antibiotics at least until 24 hours afebrile.

Continue for 7-10 days or change based on culture and sensitivity.

4. Evidence supported therapy- Change antibiotics based on culture results.
5. Surgical treatment:
 - a. Evacuation of any products of conception.
 - b. Evacuation of any foreign body e.g. forgotten gauze.
 - c. Release of sutures on an infected wound for pus to drain.
 - d. Surgical debridement+/- secondary wound closure with non-absorbable, non-woven/monofilament suture material.
 - e. Incision and drainage of a local collection on skin or subcutaneous tissue.
 - f. Exploratory laparotomy with washout of peritoneal collection.
6. Fetal surveillance- using available strategies-fetal heart auscultation (fetoscope), Cardiotocography (CTG), Obstetric Ultrasound (BPP).
7. Delivery plan:
 - a. If at term, initiate delivery plan.
 - b. If remote from term, stabilize the mother and manage the source of infection as you discuss with team on pregnancy follow up based on initial response.

Summary of the antibiotics of choice

Mild and Community acquired infections:	Sever infections/2nd line regimen
- Cefuroxime 1.5g IV TDS + Gentamycin 4mg/kg STAT (maintenance 80mg TDS)+ Metronidazole 500mg TDS. OR *Clindamycin 900mg TDS+. Gentamycin 80mg TDS+. Metronidazole 500mg TDS. OR - Amoxicillin/clavulinate 1.2g TDS or Amikacin 15mg/kg OD+ Metronidazole 500mg TDS. - to review antibiotics based on latest AMR guidelines.	- Piperacillin-tazobactam 4.5g TDS. OR - Meropenem 1g TDS. PLUS - Clindamycin 900mg TDS. If MRSA, - Add Vancomycin 25-30mg/kg STAT. Postpartum Sepsis: - Clindamycin 900mg TDS. - Gentamycin 4-7mg/kg STAT then 80mg TDS.

*Allergic to penicillin



Algorithm for management of maternal sepsis

Suspected maternal sepsis:

Symptoms	Signs
Fevers &/or chills.	Tachycardia>110b/min.
Smelly PV discharge.	Tachypnea>22br/min.
Abdominal pain.	Hypoxia <90%.
Altered mentation.	Hypotension-sbp<90mmhg.



'Golden hour' approach:

- 2 Large bore cannula.
- Antibiotics-see regimens above.
- Antipyretics.
- Fluid resuscitation-Normal saline or ringers lactate 30mls/kg.
- Collect samples for septic screen-haemogram, CRP, procalcitonin, Blood culture, urine culture.
- Once stable, consider imaging as appropriate- pelvic U/S, abdominal X-ray, CXR.



Antibiotic regimen

FIRST LINE ANTIBIOTICS (5-7 day course).

1. Cefuroxime 1.5g IV TID.
+Gentamycin 80 TID+ Metronidazole 500mg TID IV.*

OR

2. Clindamycin 900mg TID+.
Gentamycin 80mg TID+.
Metronidazole 500mg TID.*

OR

3. Clindamycin 600mg IV TDS.
+Gentamycin 4-7mg/kg STAT then 80mg TDS.**



Subsequent principles

Multidisciplinary teams- intensivists, infectious disease specialists, obstetricians.

Critical care/HDU where indicated
Exploratory laparotomy if indicated
If Pregnant:

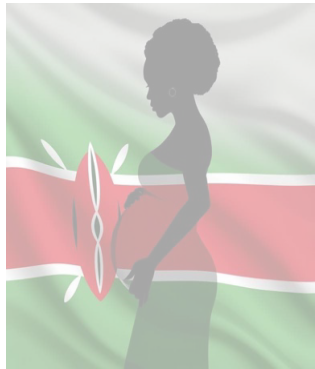
- Fetal surveillance.
- Decision on timing and mode of delivery.

SECOND LINE ANTIBIOTICS

REGIMENT: CONSULT WITH OR REFER TO LEVEL 4 AND ABOVE.

Strategies for prevention of maternal sepsis

- Access to clean water and sanitation.
- Access to quality obstetric care.
- Prevention and management of risk factors such as anemia, HIV, diabetes mellitus and malnutrition.
- Safe abortion practices.
- Correct and timely use of antibiotics where necessary.



6. Septic Abortion

Definition:

This is abortion characterized by infection of the products of conception, infection of the lining of the uterus and may lead to generalized infection.

Common signs and symptoms of septic abortion

1. Abdominal and pelvic pain.
2. Purulent vaginal discharge.
3. Vaginal bleeding.
4. Fevers and chills.

Suspected maternal sepsis:

Symptoms	Signs
PV bleeding.	Uterus tender.
Foul smelling PV blood/discharge.	Tachycardia >100b/min.
Abdominal/pelvic pains.	Tachypnea >20b/min.
Fevers, chills, rigors.	SBP <90mmHg.
Others: headaches, nausea, vomiting, diaphoresis.	Temperature >38 °C <36°C.

Severe post-abortion infection

Criteria for the systemic inflammatory response syndrome.

The systemic inflammatory response syndrome (SIRS) is clinically recognized by the presence of two or more of the following:

Temperature > 38C or <36C.

Heart rate >90 beats/min.

Respiratory rate >20 breaths/min or PaCO₂ <32mmHg.

WBC >12,000 cells/mm³, <4000 cells/mm³, or >20 percent immature (band) forms.

Risk factors to consider in diagnostic evaluation:

1. History of any abortion/miscarriage.
2. Uterine instrumentation.
3. Prolonged per vaginal bleeding in the setting of a pregnancy event.

Investigations for septic abortion

Laboratory tests:

1. Complete blood count- with differential counts- see table above.
2. Kidney function tests- urea, creatinine levels.
3. Grouping and cross match –in cases where patient is heavily bleeding, is anemic or may require surgical evacuation.
4. Lactate levels.
5. Blood gas analysis.
6. Urinalysis and urine culture to assess for infection.
7. Blood cultures.
8. Screening for sexually transmitted infections- in high risk patients-gonorrhea, chlamydia, trichomoniasis, HIV and syphilis.

Imaging tests

1. Pelvic Ultrasonography:
 - a. Check for retained products of conception.
 - b. Check for uterus size and volume.
 - c. Pelvic/tubo-ovarian collection.
2. CT scan abdomen- in case of gas forming organisms e.g. clostridia species or gas due to necrotizing fasciitis.

Management

Principles of management

1. Timely recognition and diagnosis.
2. Assess need for early, timely and appropriate referral.
3. Clinical stability and restoration of physiological function.
4. Circulatory support through intravenous fluid resuscitation.
5. Broad spectrum antibiotic.
6. Pain management.
7. Infection source control .
8. Other comprehensive post-abortion care.
 - a. Counseling.
 - b. Emergency treatment of incomplete abortion- septic abortion will need surgical evacuation.
 - c. Family planning and contraception services.
 - d. Linkage with reproductive and other health services such as screening.
 - e. Community and service provider partnership.

Clinically unstable patient (features of septic shock as above)

1. Haemodynamic stabilization.
 - a. IV fluids resuscitation- crystalloids e.g. Normal saline, Ringers lactate 30mls/kg.
 - b. Oxygen if saturations <90% on room air using nasal prongs, face mask or non-rebreather mask.
 - i. 4L/min via nasal prongs.
 - ii. 10-15L/min when using the Non-Rebreather mask.
2. Initiation of IV antibiotics- see table below.
3. Early surgical evacuation using MVA.
4. Some cases with uterine perforation and intra-abdominal sepsis may warrant laparotomy +/- hysterectomy.
5. Cross-discipline consultation with infectious disease specialists and critical care team.
6. Admit to the critical care unit.

Stable patient

1. Pain control:
 - a. Paracetamol 1g 8 hourly for 2-3 days.
 - b. NSAIDS such as Ibuprofen 400mg 12 hourly 2-3 days.
 - c. or any other available analgesics.
2. IV fluids as appropriate:
 - a. Crystalloids (0.9% saline, Ringer's lactate) at 30mls/kg and monitoring.
3. Empirical antibiotics- see below.

Common antibiotic regimens for Septic abortion

IV Antibiotics regimen (options)	Oral antibiotics (after afebrile for 24hrs) for 10-14 days
First line regimen: until afebrile for 24 hours. - Ampicillin 2g IV every 4 hours+ Gentamycin 5mg/kg OD+ Metronidazole 500mg TDS. OR - Ampicillin 2g IV every 4 hours + Gentamycin 5mg/kg OD + Clindamycin 900mg TDS. Second line regimen (Consult with or refer to level 4 and above) - until afebrile for 24 hours. - Piperacillin-tazobactam 4.5g IV TDS with or without Vancomycin. OR - Meropenem/Imipenem + Cilastatin 500mg QID. OR - Cefepime or Ceftazidime 2g IV TDS + Metronidazole 500mg TDS. OR - Levofloxacin 500mg IV OD + Metronidazole 500mg TDS.	- Doxycycline 100mg BD. - Metronidazole 400mg TDS.

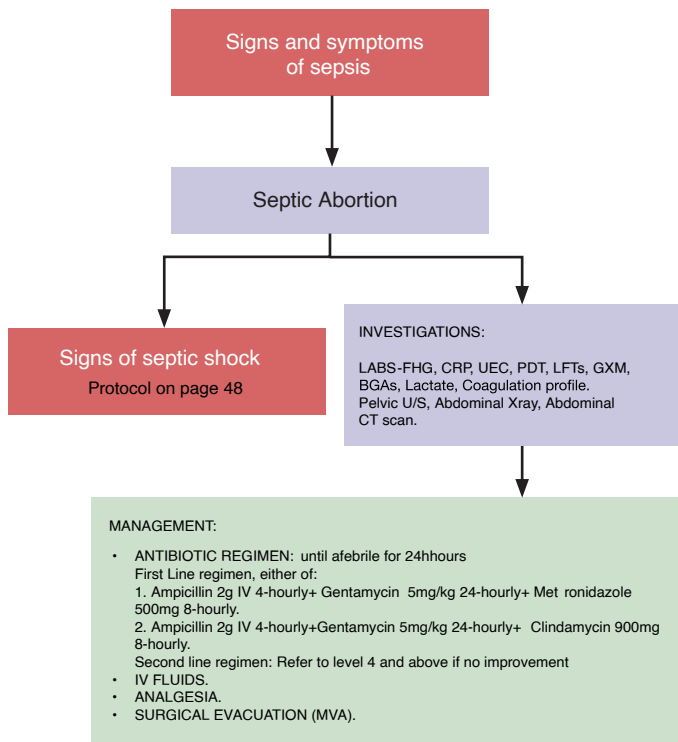
Surgical Evacuation in septic abortion:

1. MVA be commenced as soon as commencement of antibiotics.

Indications for Laparotomy +/- Hysterectomy:

1. Lack of clinical improvement within 48 hours despite surgical evacuation and adequate antibiotic cover.
2. Ongoing and severe intra-abdominal infection:
 - a. Persistent fevers.
 - b. Unstable vital signs with tachycardia and tachypnea.
 - c. Features of obstipation- abdominal distension, abdominal pains, not able to pass stools, vomiting.
3. Massive hemorrhage during or following surgical evacuation.
4. Significant uterine perforation as a cause of the septic abortion or during surgical evacuation.

Algorithm for management of septic abortion



Strategies for prevention of unsafe abortion

- Reduce unintended pregnancies.
- Provision of post-abortion care to prevent complications of unsafe abortion.
- Provide family planning services and offer post-abortion counseling.
- Strengthening policies that promote safe post-abortion care.
- Monitoring and evaluation of adherence to comprehensive post-abortion care services.

Reference ranges

NORMAL RANGES IN PREGNANCY

Test Category	Analyte	Unit	Range
Hematology	ESR (Westergren)	mm/h	0-20
	Hemoglobin	g/dL	Mild 10-11
	Moderate 7-10		
	Severe less than 7		
	RBC Count	$\times 10^{12}/L$	4.1-5.1
	WBC Count	$\times 10^9/L$	6.0 – 15.0
	Platelets	$\times 10^9/L$	150 – 400
	PT	s	11-13.5
	INR	ratio	0.8-1.2
	Hematocrit	%	36-46
	MCV	fl	80-100
	MCH	ng	27-33
	MCHC	g/dl	32-36
	RDW	%	11.5 -14.5
	Reticulocyte	%	0.5-2.5
	Neutrophils	%	40-70
	Lymphocytes	%	19-48
	Monocytes	%	2-10
	Eosinophil	%	0-6
	Basophil	%	0-2
	Absolute Neutrophil Count	$\times 10^9/L$	1.5-7
	Absolute Lymphocyte Count	$\times 10^9/L$	1-3.5
	Absolute Monocyte Count	$\times 10^9/L$	0.2-1
	Absolute Eosinophil Count	$\times 10^9/L$	0-8
	Absolute Basophil count	$\times 10^9/L$	0-0.2
Renal	Sodium	mmol/L	130-140
	Potassium	mmol/L	3.3-5.1
	Creatinine	$\mu\text{mol}/L$	35-80
	Urea (BUN)	mmol/L	1.0 – 4.5
	Uric Acid	$\mu\text{mol}/L$	120 - 300
Glucose	Glucose (fasting)	mmol/L	less than 5.1
	OGTT 75g	mmol/L (1hr)	10
	OGTT 75g	mmol/L (2hr)	8.5
	HBA1c	%	4-5.4

Liver function tests	Aspartate Aminotransferase (AST)	U/L	3-37
	Alanine Amino Transferase (ALT)	U/L	0-34
	Alkaline phosphatase(ALP)	U/L	40-250
	Gamma glutamyl transferase (GGT)	U/L	8-35
	Lactate dehydrogenase (LDH)	U/L	140 -280
	Albumin	g/l	25-37
	Total Protein	g/l	64-83
	Total Bilirubin	$\mu\text{mol/L}$	5-21
	Direct (Conjugated Bilirubin)	$\mu\text{mol/L}$	0-5
Urinalysis (Dipstick)	RBC	Positive/Negative	Negative
	Protein	Positive/Negative	Negative
	Glucose	Positive/Negative	Negative
	Pus cells	Positive/Negative	Negative

Maternal Health Job Aids



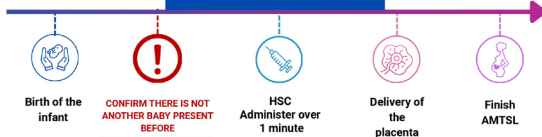
Ministry of Health

Heat Stable Carbetocin (HSC)

HSC use for prevention of PPH

UTEROTONIC: HSC TO BE ADMINISTERED IMMEDIATELY AFTER BIRTH OF INFANT

THIRD STAGE OF LABOUR



Administration guideline

Administer a 1 mL dose (equivalent to 100 micrograms) either intramuscularly or as a slow intravenous injection over one minute directly through the IV port.

DO NOT inject HSC into the IV fluid bag



- The product is supplied in packs of ten 1 mL ampoules, each containing 100 micrograms.
- One ampoule provides the recommended dose and is administered undiluted by IM or slow IV injection.

Storage



Does not require refrigeration



Keep away from direct light



Use before expiry date

DO NOT USE HSC in these scenarios:

- For induction or augmentation of labour
- At any time during pregnancy or labour before delivery of the baby
- In women with severe cardiovascular disease
- In women who are hypersensitive to carbetocin, oxytocin, or any listed excipients in the formulation
- In women with hepatic or renal impairment
- In women with epilepsy





Post Partum Hemorrhage

Prevention and diagnosis

Provide respectful maternal care
Ensure Asepsis to prevent infections
Prepare for birth - Check equipment & medications

Prepare for birth
Check equipment and medications

Assess & record

- Every 15 mins
- Assess uterine tone
- Assess blood loss and flow
- Assess blood pressure and vitals
- Encourage breast feeding



Be alert for PPH

- ≥ 300 ml and abnormal vital signs
- OR
- Blood loss ≥ 500 ml
- OR
- Clinical judgment
- Blood loss of 1000mls during cesarean section delivery.

Give uterotonic within 1 minute of delivery of baby
Carbetocin - 100mg 1U or IV
OR Oxytocin 10 IU IM or IV
OR Misoprostol 600 µg PO

Measure &
record
every 15
minutes

Measure blood loss with calibrated drape

Delayed cord clamping (1-3 minutes)

Perform Controlled Cord Traction (CCT)

Assess for complete delivery of placenta

OUT

Confirm complete placenta

Assess for tears & repair

NOT OUT

Empty bladder, repeat CCT

NOT OUT After 30 mins

Manual removal of placenta

Concerns?

Start advanced care → PPH Start Bundle → Escalate if bleeding persists



Post Partum Hemorrhage

Treatment

Provide respectful maternal care
Ensure Asepsis to prevent infections

PPH Diagnosed
≥300ml and vital signs abnormal
Blood loss ≥500ml
Clinical signs & symptoms

ACTIVATE PPH BUNDLE

E

Early Detection
and Trigger
Criteria

M

Massage
uterus

O

Oxytocin 10 IU
and Misoprostol
800mcg

T

Tranexamic
Acid

IV

IV
Fluids

E

Examine &
address cause.
Tone: Tissue
Trauma: Thrombosis

STILL BLEEDING?

NO

YES

Surgical

Non Surgical

- Communicate
- Assign roles

Assess need for:
1. NASG
2. Referral/Surgery
3. Manage shock

- Repeat tranexamic
- Additional uterotonic → 20 IU in 1000ml IV
- Start second IV
- Continuous massage
- Empty bladder
- Clear uterus and cervix

Other interventions:

- Safe Blood Transfusion
- Repair of tears
- Bimanual compression
- Balloon tamponade
- Aortic compression

Assess and record

- Monitor every 15 mins to 1 hour
- Then every 30 mins x 4

NO

STILL BLEEDING?

YES

Surgical Interventions

- Uterine compression sutures:
B-lynn suture, Hayman sutures.
- Stepwise Uterine de-vascularization
- Subtotal/Total Abdominal Hysterectomy



Ministry of Health

Maternal Resuscitation

Attempt resuscitation for 4 minutes after which consider perimortem CS for eligible candidates.

Maternal unresponsiveness

1. Shout for help! (colour codes)
2. Note the time
3. Activate Emergency team
4. Get the crash cart immediately!

Maternal cardiac arrest

1. Start CPR immediately without delay
2. Think of the treatable causes (See below)

Basic Life support



High Quality Chest Compression

- Perform over the Center of the Lower Half of the Sternum
- Rate: 100 Per Min
- Depth: At least 2 inches
- Allow Full Chest Recoil
- Chest Compression to be Maintained till ROSC



Manual Left Uterine Displacement

- Mandatory for Gestational Age > 20 wks or Uterus Above Umbilicus
- Bimanual Displacement by a Dedicated Person Either Pulling from Left or Pushing From Right
- Should Be Maintained Throughout without any Interruption
- This Maneuver is a Life Saver



Airway & parturient Position

- Continue Compression only CPR till Help Arrives
- Bag Mask Ventilation with 100% O2 using Two Hands is More Effective
- Keep Parturient in Supine Position on a Firm Hard
- Surface on Back
- Avoid Left Lateral Tilt, as it Interferes with High quality CPR
- 30:2 Compression Ventilation Ratio

Perimortem cesaarean section



Consider Early

- IF ROSC is Not Achieved within 5 Minutes of Arrest
- Strongly Recommended for Gestational Age > 20 wks
- Lower the Arrest-Delivery Interval
- The Lower the Arrest-Delivery Interval the Higher the Survival of the Infant

Do it fast

- To be Performed at the Site of Arrest Without Shifting to the Operation Room
- Abbreviated Antisepsis
- Use Scalpel & Clamps
- Maintain High Quality Chest Compressions & Manual Left Uterine Displacement Throughout

Post Arrest Care

- Be Careful with Uterotonics
- Achieve Swift Haemostasis and Surgical Closure
- Organ Support and Protective Strategies should be Initiated
- Targeted Temperature Management must be Planned

Advanced life support

Advanced Airway

- Obstetric Airway is Difficult
- Secure Early to Minimize the Risk of Aspiration & Hypoxic Brain Injury
- Use a Smaller Size ET tube 6 or 7 mm
- Secure the Airway with SGA Device in Case of Difficult/Failed Intubation
- Ventilate 8-10 Breaths per Min with 100% Oxygen



Possible causes

General

- A: Anaphylaxis
- B: Bleeding
- C: Cardiac
- D: Drugs
- E: Embolism

Obstetric

- U: Uterine
- A: ATE
- U: Uterine Rupture
- P: Placental
- P: Pulmonary
- S: Sepsis

Emergency Cart

- Adrenaline
- Amiodarone
- 20% Intralipid
- Calcium Gluconate
- Magnesium Sulphate
- Naloxone
- Sodium Bicarbonate
- Automated External Defibrillator



Disclaimer

Errors and Omissions: While every effort has been made to ensure that the information contained herein is accurate and up-to-date at the time of publication, the authors and publishers assume no responsibility for any errors, inaccuracies, or omissions.

Dynamic Nature of Medicine: Medical knowledge and standards of practice are constantly evolving. Readers are advised to confirm the information with other reliable sources and consult a qualified healthcare provider regarding any specific medical condition.

